ModelIt!

Teacher's Guide

Superbugs: Evolution You Can See

Why Antibiotics Are Losing the War Against Bacteria

Quick Reference

Grade Level	Grade 8
NGSS Standard	MS-LS4-4, MS-LS4-6
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: MS-LS4-4, MS-LS4-6

Construct an explanation based on evidence that describes how genetic variations of traits in a population increase some individuals' probability of surviving and reproducing in a specific environment. Use mathematical representations to support explanations of how natural selection may lead to increases and decreases of specific traits in populations over time.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how natural selection drives antibiotic resistance.
Disciplinary Core Idea	**LS4.B Natural Selection** — Natural selection leads to the predominance of certain traits in a population and the suppression of others.
Crosscutting Concept	**Cause and Effect** — Students identify that antibiotic overuse CAUSES the selection pressure that leads to the EFFECT of resistant populations.

CAST Testing Alignment

MULTIPLE CHOICE:

A population of bacteria is exposed to an antibiotic. After 10 generations, 90% of the population is resistant. Which best explains this change?

CONSTRUCTED RESPONSE:

Using the concept of natural selection, explain why antibiotic-resistant bacteria have become more common in hospitals over the past 50 years.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Antibiotic resistance is one of the most urgent public health crises of the 21st century. The WHO has called it a 'global health emergency.' Understanding the evolutionary mechanisms behind resistance is essential for developing strategies to combat it.

How Resistance Develops

Bacterial populations contain natural genetic variation due to random mutations. Some bacteria happen to carry genes that provide resistance to antibiotics. When antibiotics are applied, susceptible bacteria die while resistant ones survive and reproduce, passing on their resistance genes. This is natural selection in action.

The Overuse Problem

When antibiotics are overprescribed, used for viral infections (where they don't work), or not completed as prescribed, they create stronger selective pressures. Agricultural use of antibiotics in livestock accounts for roughly 70% of all antibiotic use in the U.S.

MRSA and Other Superbugs

Methicillin-resistant Staphylococcus aureus (MRSA) is one of the most well-known superbugs. It evolved resistance to multiple antibiotics and kills approximately 20,000 Americans per year. Other concerning superbugs include CRE, VRE, and drug-resistant tuberculosis.

Mathematical Modeling

Resistance evolution can be modeled mathematically using population genetics equations. The rate of resistance increase depends on: initial frequency of resistant alleles, strength of selective pressure (antibiotic use), fitness cost of resistance, and population size.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System
Students identify antibiotic use, mutation rate, resistant bacteria percentage, and treatment effectiveness as system components.

E — Establish Relationships
Students determine that antibiotic use positively affects resistant bacteria percentage (by selecting for them) but negatively affects treatment effectiveness.

V — Visualize & Model
Students build the resistance evolution model showing how selection pressure drives population change.

E — Evaluate Outcomes
Students run overuse vs. responsible use scenarios and compare the rate of resistance evolution.

R — Revise & Extend
Students add horizontal gene transfer or immune system components to make the model more realistic.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"Have you ever taken antibiotics? What if they stopped working?"*

Watch For

Share a real news headline about superbugs to build urgency.

Slide 2: Learning Objectives

3 min

Say: *"Today we're modeling evolution happening in real time — right now, in hospitals."*

Watch For

Have students read objectives. Pre-teach 'mutation' vs. 'natural selection.'

Slide 3: Big Question

3 min

Say: *"This isn't science fiction. The WHO calls this a global health emergency."*

Watch For

Poll: How many students have taken antibiotics? Why were they prescribed?

Slide 4: LEVER Framework

2 min

Say: *"We'll use modeling to understand how evolution creates superbugs."*

Watch For

Briefly preview each LEVER step.

Slide 5: Activity 1: Components

8 min

Say: *"What are the key parts of this system? What can we control vs. what happens on its own?"*

Watch For

Guide sorting; emphasize that mutation is RANDOM but selection is NOT.

Slide 6: Activity 2: Connections

8 min

Say: *"Here's the key insight: antibiotics don't CAUSE resistance — they SELECT for it."*

Watch For

Demonstrate positive/negative relationships. Focus on the counterintuitive idea.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's see what happens when we overuse antibiotics vs. use them responsibly."*

Watch For

Have students predict before running. Compare overuse vs. smart use graphs.

Slide 8: Discoveries

5 min

Say: *"What did our model reveal? How does this change how you think about antibiotics?"*

Watch For

Lead whole-class discussion. Connect to real MRSA data.

Slide 9: STEM Challenge

12 min

Say: *"You're now consultants for a hospital. Design a policy to fight superbugs."*

Watch For

Distribute materials. Circulate and ask probing questions about evidence.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Antibiotic Use, Mutation Rate
- **INTERNAL:** Resistant Bacteria, Treatment Effectiveness

Relationships

- **Antibiotic Use to Resistant Bacteria:** POSITIVE (+) — More antibiotic use creates stronger selective pressure, allowing resistant bacteria to outcompete susceptible ones.
- **Mutation Rate to Resistant Bacteria:** POSITIVE (+) — Higher mutation rates create more genetic variation, increasing the chance of resistance genes appearing.
- **Resistant Bacteria to Treatment Effectiveness:** NEGATIVE (-) — As the percentage of resistant bacteria increases, antibiotics become less effective at treating infections.

Key Discoveries

- Random mutations create resistant bacteria BEFORE antibiotics are even used
- Antibiotics don't cause resistance — they SELECT for bacteria that already have it
- Overuse accelerates the evolution of resistant populations dramatically
- Reducing antibiotic use slows resistance evolution but doesn't reverse it instantly

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Antibiotics make bacteria resistant"	Antibiotics don't CREATE resistance — they SELECT for bacteria that already have resistance due to random mutations. The resistance genes existed before the antibiotic was ever used.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Evolution takes millions of years"	Bacteria can evolve noticeable resistance in days or weeks because they reproduce so quickly (every 20 minutes). Evolution by natural selection can be observed in real time.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"If I feel better, I can stop antibiotics"	Stopping early kills the weakest bacteria but leaves the most resistant ones alive to reproduce. Completing the full course ensures even partially-resistant bacteria are killed.